PRODUCT REQUIREMENTS DOCUMENT (PRD) • B2B VERTICAL AI AGENT

BetriebsPilot AI: Autonomous Back-Office Agent for Legacy ERPs

Zero-Touch Order-to-Cash, Invoicing, Returns (RMA), and Inventory Operations for Small German Suppliers running SAP Business One, Sage 100, proALPHA & abas

Document Version: v1.0	**Target Market:** DACH Small Suppliers (10–150 Employees)	**Architecture:** Hybrid Cloud / 100% On-Premise Local LLM	**Status:** Ready for Engineering Execution

1. Executive Summary & Product Vision

The Vision: Empower small and mid-sized German suppliers (*Mittelstand Großhandel & Zulieferer*) to run a complete, high-volume commercial back-office on legacy ERP systems (**SAP Business One, Sage 100, proALPHA, abas, SelectLine**) with a 1–2 person supervision team instead of 5–10 full-time back-office clerks (*Sachbearbeiter*).

Product Definition: BetriebsPilot AI is an autonomous, multi-channel AI Operations Agent connected to legacy ERPs via a standardized **Model Context Protocol (MCP) Bridge**. Unlike simple OCR tools that only read PDFs, BetriebsPilot AI executes the entire **Order-to-Cash (O2C) and Inventory lifecycle**: ingesting orders from PDFs, emails, WhatsApp, and SMS; generating delivery notes and compliant e-invoices (*ZUGFeRD / XRechnung*); processing customer returns (*Retouren / RMA*) and credit notes (*Gutschriften*); and autonomously managing stock reservations and supplier reorder alerts.

85% – 92%	**< 15 Sec**	**≥ 96%**	**100% GoBD**
REDUCTION IN MANUAL ERP DATA ENTRY TIME	ORDER-TO-ERP PROCESSING TURNAROUND	SKU & UOM RESOLUTION ACCURACY	& EN 16931 E-RECHNUNG COMPLIANCE

2. Problem Statement & Market Context

Small B2B suppliers in Germany (selling 5,000 to 50,000 SKUs such as fasteners, seals, electrical components, HVAC supplies, or packaging) face an existential operational squeeze:

- **Severe Skilled Labor Shortage (*Fachkräftemangel*):** Experienced clerks who know arcane ERP transaction codes and product catalogs are retiring. Small suppliers cannot afford or recruit large administrative teams to manually type orders, invoices, and returns into clunky 1990s desktop interfaces.
- **Multi-Channel Unstructured Chaos:** Customers no longer order via a single channel. Orders, return complaints, and stock inquiries arrive via **PDF attachments, messy emails, WhatsApp messages (with photos of broken parts), and SMS** from field technicians.
- **Fragmented Back-Office Workflows:** Even if an order is typed in, clerks spend hours manually converting orders into delivery notes (*Lieferscheine*) and legally mandated electronic invoices (*E-Rechnung*), handling defective/wrong-item returns (*Retourenabwicklung*), and checking stock levels across warehouses.
- **ERP Migration Paralysis:** Replacing SAP B1, Sage 100, or proALPHA costs €150k–€500k+ and disrupts business for 18 months. Suppliers need an intelligent automation layer that operates their *existing* ERP autonomously.

3. Target Users & Buyer Personas

Persona	Current Daily Pain	How BetriebsPilot AI Transforms Their Role
Geschäftsführer (Owner / Managing Director of a 15–80 person supplier)	Paying €250k+/year for 5 clerks doing repetitive data entry in Sage/SAP; orders delayed when staff are sick or on vacation.	Runs the entire commercial back-office with 1–2 exception supervisors; scales order volume 3x without adding headcount.
Allround-Sachbearbeiter (Lean Back-Office Operator)	Jumping between Outlook, WhatsApp Web, Excel, and 12 open ERP windows to enter orders, create invoices, and book returns.	Reviews a single unified **Approval Inbox**; approves low-confidence exceptions with 1 click while 80%+ of routine tasks run hands-free.
Lagerleiter / **Einkauf** (Warehouse & Purchasing Lead)	Stockouts discovered too late; manual counting of returned goods; tedious supplier reorder emails.	Receives instant WhatsApp/Teams stock alerts, automated return inspection tickets, and pre-drafted supplier Purchase Orders (*Bestellvorschläge*).

4. Core Autonomous Agent Capabilities (The 4 Functional Pillars)

Pillar 1: Multi-Channel Sales Order Agent (PDF, Email, WhatsApp, SMS)

- **Omnichannel Ingestion:** Monitors shared IMAP/Exchange inboxes (`bestellung@`), Twilio/WhatsApp Business webhooks (including voice notes and photos of part labels), and SMS gateways.
- **3-Tier Cascade SKU Resolution:** Matches vague descriptions (*"50x O-Ring 22mm from March quote"* or customer internal SKU `44012`) via (1) Customer-Material Cross-Reference Table, (2) Customer 24-month Order/Quote History, and (3) Hybrid Search (BM25 + Dense Vector + Cross-Encoder Reranker) over the ERP *Artikelstamm*.
- **Deterministic Packaging & Pricing Math:** Converts order units to ERP base units (e.g., $10 \text{ Kartons} \times 12 = 120 \text{ Dosen}$), verifies customer-specific tier pricing/discounts, and writes the validated Sales Order (*Kundenauftrag*) into the ERP.

Pillar 2: Autonomous Invoicing & E-Rechnung Agent (Order-to-Cash)

- **Automated Document Chain (*Belegfluss*):** Once warehouse dispatch confirms shipment (or upon instant approval for stock items), the agent automatically triggers the ERP to convert the *Kundenauftrag* → *Lieferschein* (Delivery Note) → *Ausgangsrechnung* (Sales Invoice).
- **German E-Rechnung Mandate Compliance:** Automatically generates EN 16931 compliant **ZUGFeRD 2.3 (hybrid PDF/A-3 + XML)** and **XRechnung** files, validates tax rates (19% USt, 0% intra-EU reverse charge with valid *USt-IdNr.*), and dispatches them to the customer's billing email.
- **3-Way Matching & Payment Reconciliation:** Matches incoming supplier invoices against open Purchase Orders and Goods Receipts, exporting clean accounting batches to **DATEV**.

Pillar 3: Returns, Claims & Credit Note Agent (Retouren / RMA Autopilot)

- **Multi-Modal Return Intake:** When a customer sends an email or WhatsApp message saying *"20 pieces of M8 bolts arrived damaged / wrong thread size"* (with an attached photo), the agent identifies the original ERP invoice and delivery note automatically.

- **Policy & Reason Classification:** Classifies the return reason (*Falschlieferung, Transportschaden, Kulanzrücknahme*), checks if the item is within the return window, and calculates restocking fees if applicable.
- **RMA & Credit Note Execution:** Automatically generates an RMA return label (DHL/UPS API), creates a Return Delivery slip (*Retourenschein*) in the ERP, triggers replacement shipment or drafts a legally linked Credit Note (*Gutschrift / Rechnungskorrektur*) awaiting 1-click human sign-off.

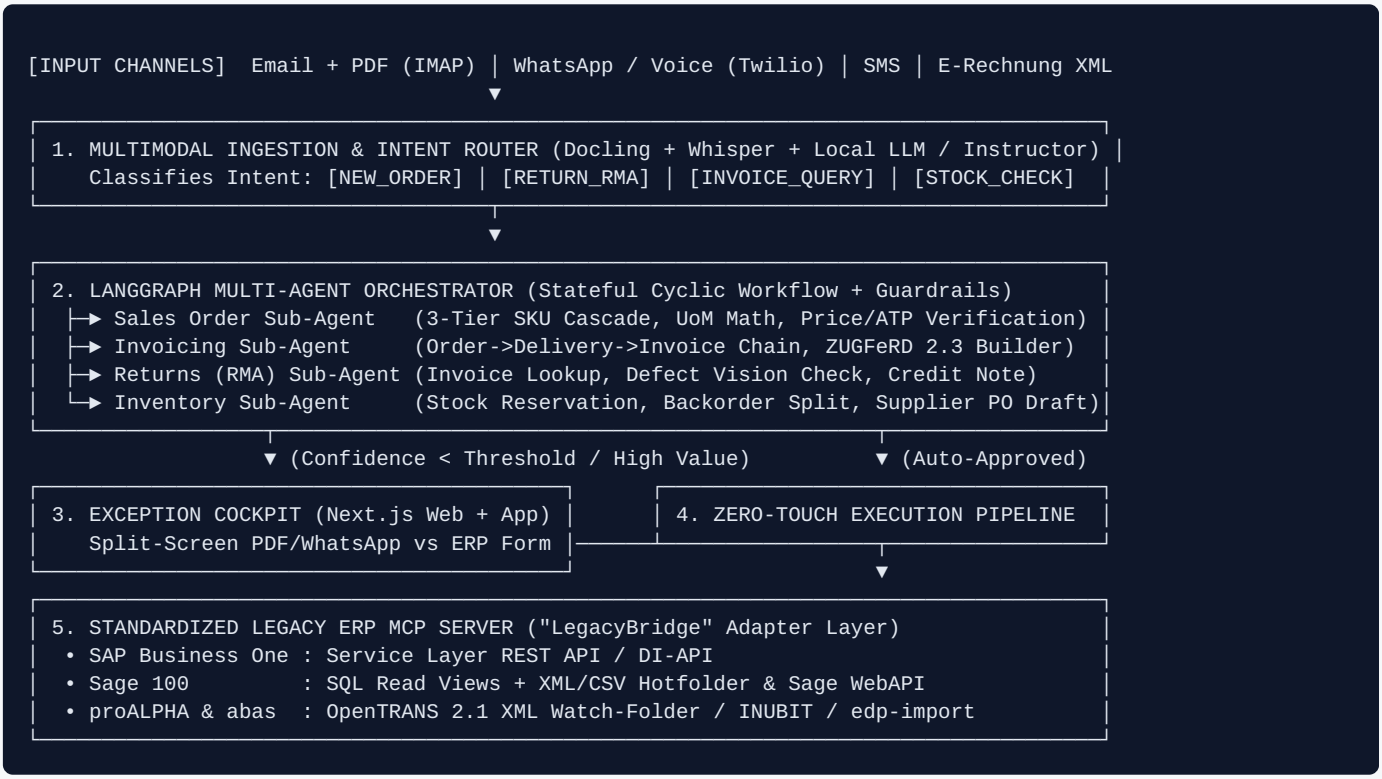
Pillar 4: Inventory & Replenishment Agent (Bestands- & Einkaufs-Agent)

- **Real-Time ATP (Available-to-Promise) Check:** Before confirming any sales order, the agent checks physical stock minus existing reservations ($ATP = Stock_{phys} - Stock_{reserved} + Stock_{incoming}$).
- **Automated Backorder & Substitute Handling:** If an item is out of stock, the agent automatically identifies technical drop-in substitutes in the catalog or splits the order into partial delivery (*Teillieferung*) and notifies the customer.
- **Autonomous Supplier Reordering (*Bestellvorschläge*):** Monitors minimum stock thresholds (*Meldebestand*), calculates optimal reorder quantities based on lead times, and drafts Purchase Orders (*Lieferantenbestellungen*) in the ERP.

5. Prioritized Feature Matrix & Acceptance Criteria

ID / Prio	Module & Feature	Functional Requirement	Acceptance Criteria (Definition of Done)
P0 (MVP) FR-01	**Omnichannel Order Ingestion**	Parse PDFs (Docling), Email bodies, and WhatsApp/SMS text/audio into structured Pydantic schemas.	Extracts 100% of line items on 50-sample German test set in < 6s using local Qwen2.5-7B/14B model.
P0 (MVP) FR-02	**3-Tier Cascade SKU & UoM Resolver**	Resolve customer SKUs, historical references, and fuzzy German descriptions; convert packaging units.	Recall@1 ≥ 90%, Recall@3 ≥ 98% on 10k-SKU catalog; 100% deterministic packaging math accuracy.
P0 (MVP) FR-03	**Confidence Router & Split-Screen UI**	Auto-post orders with confidence ≥ 0.95 (if enabled); route < 0.95 to Split-Screen Human-in-the-Loop UI.	Every human correction automatically updates `erp_customer_sku_map` for 100% future accuracy.
P1 FR-04	**Automated Invoicing & ZUGFeRD Generator**	Trigger ERP invoice creation upon delivery confirmation and emit EN 16931 compliant ZUGFeRD XML/PDF.	Passes official German KoSIT XRechnung/ZUGFeRD schema validator with 0 errors; syncs to DATEV.
P1 FR-05	**RMA / Returns & Credit Note Agent**	Match return requests to original ERP invoices, book return stock status, and draft *Gutschrift*.	Correctly links return to original `Belegnummer` and places damaged stock in quarantine warehouse.
P2 FR-06	**Inventory & Auto-Replenishment Agent**	Check ATP stock during order entry; auto-draft supplier POs when stock drops below *Meldebestand*.	Zero unflagged overselling; generates valid supplier PO draft when $ATP < SafetyStock$.

6. System Architecture & Legacy ERP Integration Strategy



7. Autonomy Tiers, Safety Guardrails & German Compliance

> **Critical Design Rule for German ERPs: Deterministic Validation + GoBD Auditability**
>
> In Germany, tax and commercial regulations (**GoBD**) prohibit unverified "black-box" mutations to financial records. The LLM **never** fabricates SKU strings, tax rates, or prices—it only invokes deterministic MCP tools and outputs structured proposals that pass strict schema and business-rule validators before touching the ERP.

Autonomy Level	Trigger Condition	Agent Action
Tier 0: Read-Only / Draft	First 14 days of customer onboarding (Shadow Mode).	Agent prepares 100% of orders, invoices, and RMAs in the staging UI; human clicks "Approve" on all.
Tier 1: Conditional Auto-Pilot	Order confidence ≥ 95%, total value < €2,500, all items in stock, known customer.	Agent writes Sales Order and Delivery Note directly to ERP; sends instant confirmation to customer.
Tier 2: Mandatory Human Gate	Credit notes (*Gutschriften*) > €100, unknown SKU (< 95% confidence), custom price override, or out-of-stock item.	Agent pauses LangGraph state and sends an interactive approval card to the clerk's Cockpit / MS Teams.

8. Phased Implementation Roadmap & Commercial Monetization

- **Phase 1 (Weeks 1–4) — Multi-Channel Order Autopilot (MVP):** Deliver Email/PDF/WhatsApp order ingestion, 3-Tier Hybrid SKU RAG, packaging unit math, Split-Screen Approval UI, and OpenTRANS XML / CSV export for Sage 100, proALPHA, and SAP B1.

- **Phase 2 (Weeks 5–8) — Inventory ATP & Automated E-Rechnung:** Add real-time warehouse stock checks during order entry, automated Delivery Note → Invoice conversion, and compliant ZUGFeRD 2.3 / XRechnung dispatch.
- **Phase 3 (Weeks 9–12) — Returns (RMA) & Autonomous Purchasing:** Launch WhatsApp/Email return processing, automated credit note drafting, and low-stock supplier PO generation.
- **SaaS Pricing Model for Small Suppliers:**
  - **Core Plan (€490/mo):** Order Autopilot + Inventory Check (up to 500 transactions/mo).
  - **Full Back-Office Autopilot (€990/mo):** Adds Automated Invoicing (ZUGFeRD), Returns/RMA Agent, WhatsApp/SMS Channel, and Supplier Reorder Proposals.
  - **On-Premise Appliance Add-On (+€300/mo):** 100% local LLM execution on a compact GPU mini-server inside the supplier's server room for strict air-gapped data privacy.